Compositional system

geometry and dynamics for,

ChemixHub HOIP-7 oxide samples

sample_00 → sample_23

Report title	Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sa
Generated	2026-05-06
Run ID	v1.1.x-final
Dataset ID	chemixhub_oxide
T (records)	24
D (carriers)	7
Engine version	cnt 2.0.4
Schema version	2.1.0
Engine signature (sha256[:32])	97748868a0381cb7b995d2ae562e5c42…
Source CSV SHA-256	9968c4a135c8b7a50cbc1af698bff4dd…
Closed data SHA-256	d123b42c416a90924eeae3d336960690… biplot
Content SHA-256 (canonical JSON)	6f18e83da444687b1149b820c61aaef6…

Reading order: geometry first, dynamics second.

```
§A  Data disclosure & processing notes
§B  Geometry : structure on the simplex
            evolution of proportions → ternaries → variation
            → correlation → scree → biplot → top-N biplot
            → balance-dendrogram → SBP table
§C  Dynamics : evolution through compositional space
            trajectories (Hs, ‖clr‖, ILR axes) → System Course Plot
            → bearing rose → bearing heatmap → ω → metric trace
            → helmsman → ring class → locks → CBS
            → pairwise divergence ranking → ω↔Hs
§D  Summary
```

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Plate 2 | §A Data Disclosure & Processing-Anomaly Review

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

```
INPUT DATA
  source file        : chemixhub_oxide_input.csv
  source SHA-256     : 9968c4a135c8b7a50cbc1af698bff4dda3dd177b146c9a5f2acf99d2ad62f62
  closed data SHA-256: d123b42c416a90924eeae3d336960690ee968034af6accc866a7d4fc74e78a0b
  records (T)        : 24
  carriers (D)       : 7
  carriers           : SiO2, TiO2, Al2O3, Fe2O3, MgO, CaO, Na2O

ORDERING
  is_temporal        : False
  ordering_method    : by-label
  caveat             : Order is treated as arbitrary; angular_velocity and energy_depth are ordering-dependent.

ZERO REPLACEMENT
  zeros replaced     : 0
  replacement δ      : 1e-15
  method             : multiplicative

UNITS LINEAGE  (schema 2.1.0)
  input_units                 : ratio
  higgins_scale_units         : neper
  units_scale_factor_to_neper : 1.0

ENGINE CONFIG ACTIVE OVERRIDES
  (none — defaults from cnt.py USER CONFIGURATION block)

ENGINE SOURCE FINGERPRINT
  cnt.py                     : c318ab82c15890b947c3fc15a6c66b55aa23f3cf94104cf9b9d5fbdb
  locked_cnt_locked.py       : 1e10cb62d6588ac9557c6a2e8436c2980f0c35b379f29c05c0377711
  combined sha256            : 97748868a0381cb7b995d2ae562e5c429988ee7fd556943b882c5fcb

DEGENERACY FLAGS
  (none)

LOCK EVENTS  (period-2 detector)
  total events       : 0

EITT BENCH-TEST RESIDUALS
  H_mean_full           : 1.6038886219729862
  M_sweep               : [{'M': 2, 'n_blocks': 12, 'H_mean_decimated': 1.60390154
  gate_pct              : 5.0
  note                  : Empirical observation of trajectory smoothness under tem

SANITY CHECKS  (computed during this report)
  ‖clr‖ non-finite count : 0
  ω    non-finite count : 0
  Hs   non-finite count : 0
  Σ CLR per row range   : [-3.33e-15, +2.44e-15]   (should be ≈ 0)
```

If any flag above is non-empty, the analysis on subsequent pages still ran but the reader should weight those plates accordingly. Determinism is preserved either way.

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Plate 3 | Evolution of Proportions (raw + cumulative)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

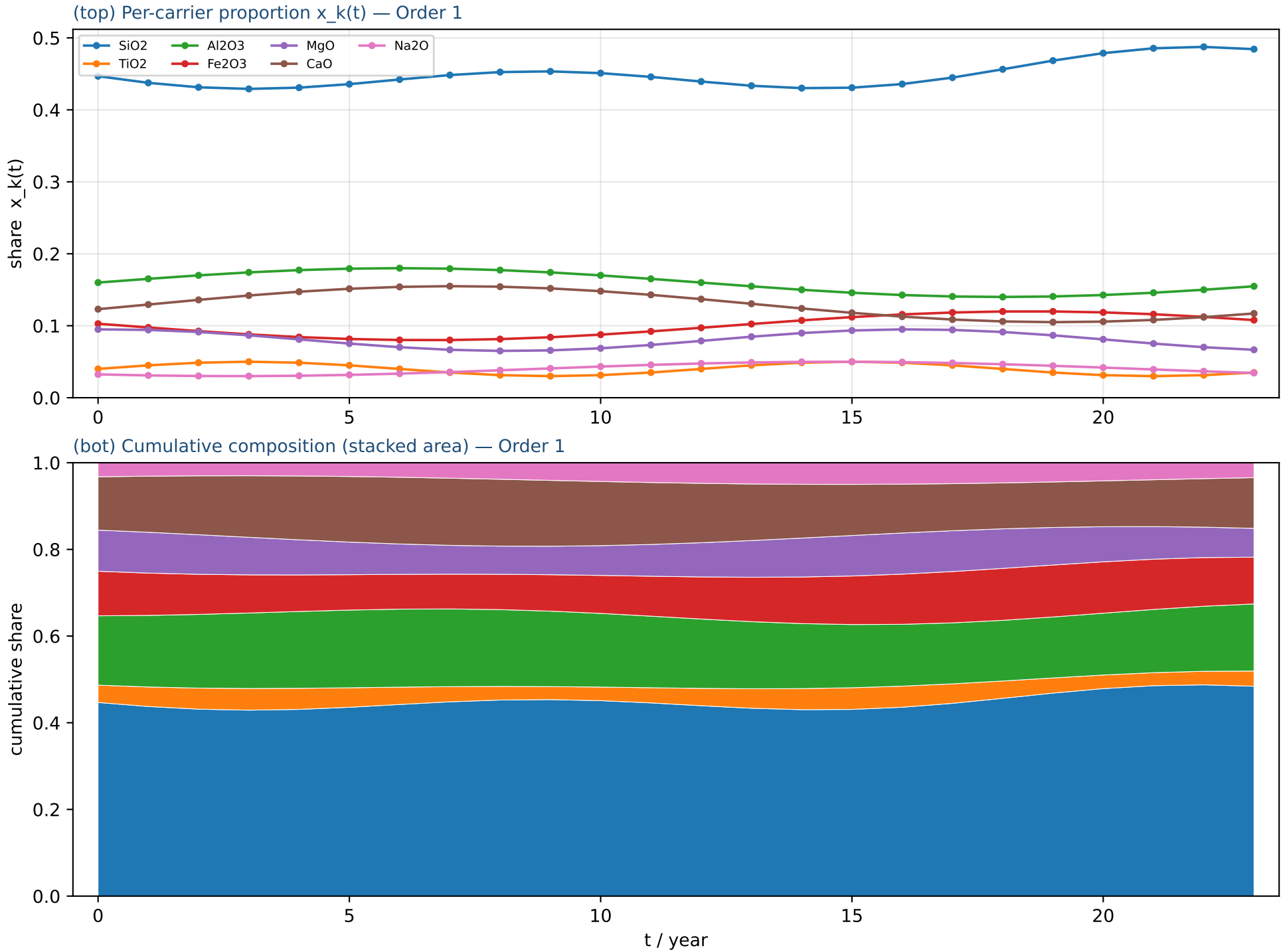

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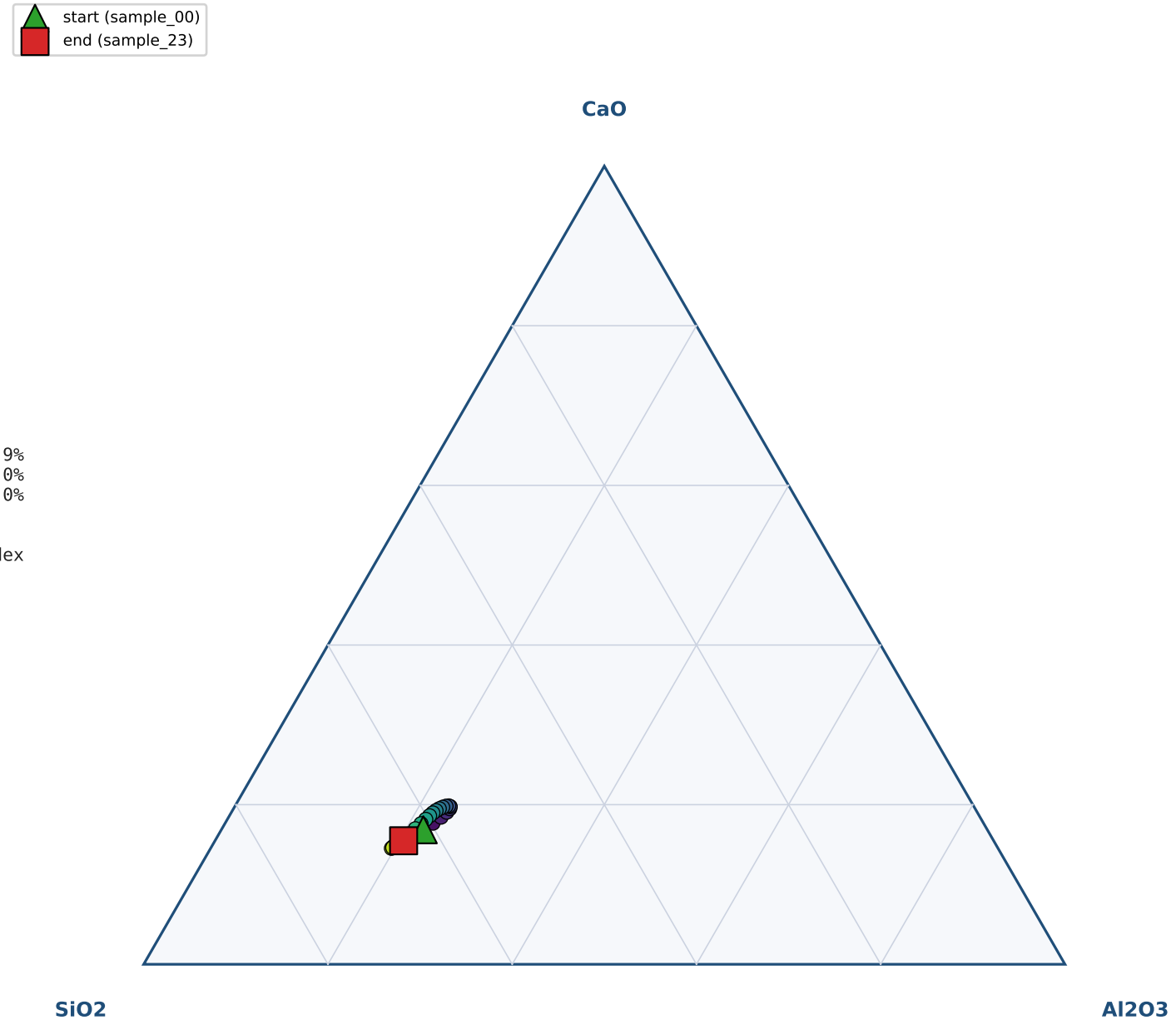
Plate 4 | Ternary Diagram (top-3 carriers)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

3-part subcomposition closed over SiO2, Al2O3, CaO
(remaining 4 carriers absorbed into closure)




```
Top-3 by mean share:
  SiO2            44.9%
  Al2O3           16.0%
  CaO             13.0%

T = 24 timesteps
colour map = time index
▲ start  ■ end
```

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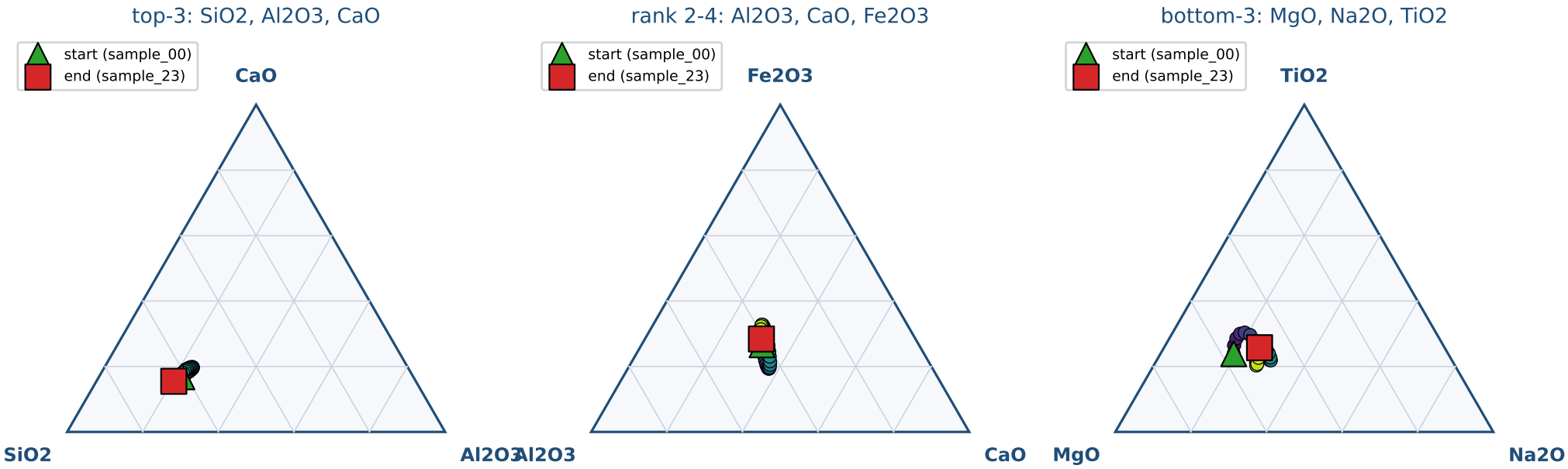
Plate 5 | Three-Part Subcompositions (3 ternaries)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



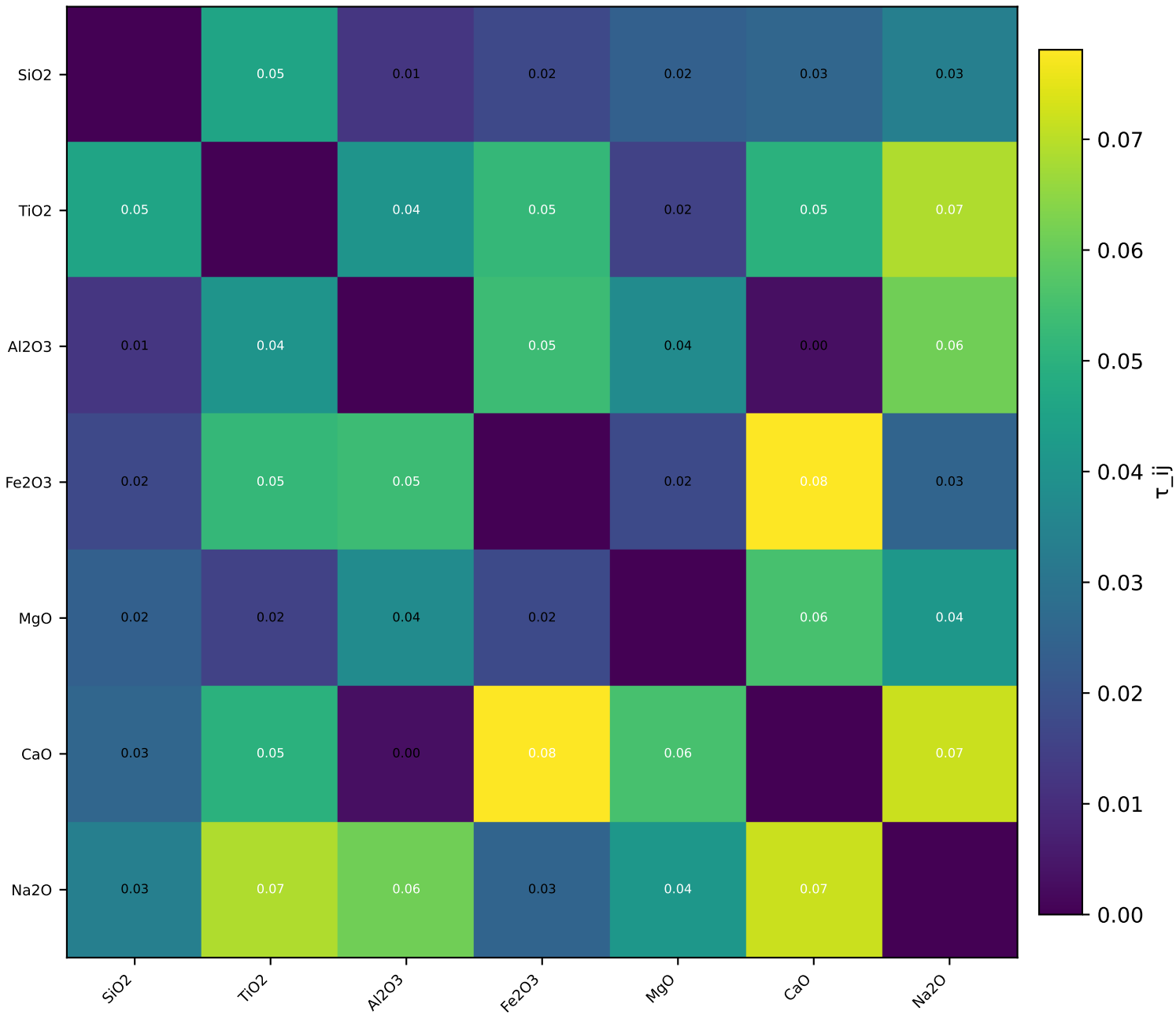
Each ternary closes the chosen 3 carriers; remaining carriers absorbed into closure. ▲ start ■ end | colour map = time index

Triplet source: default heuristic

Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 6 | Variation matrix τ_ij heatmap

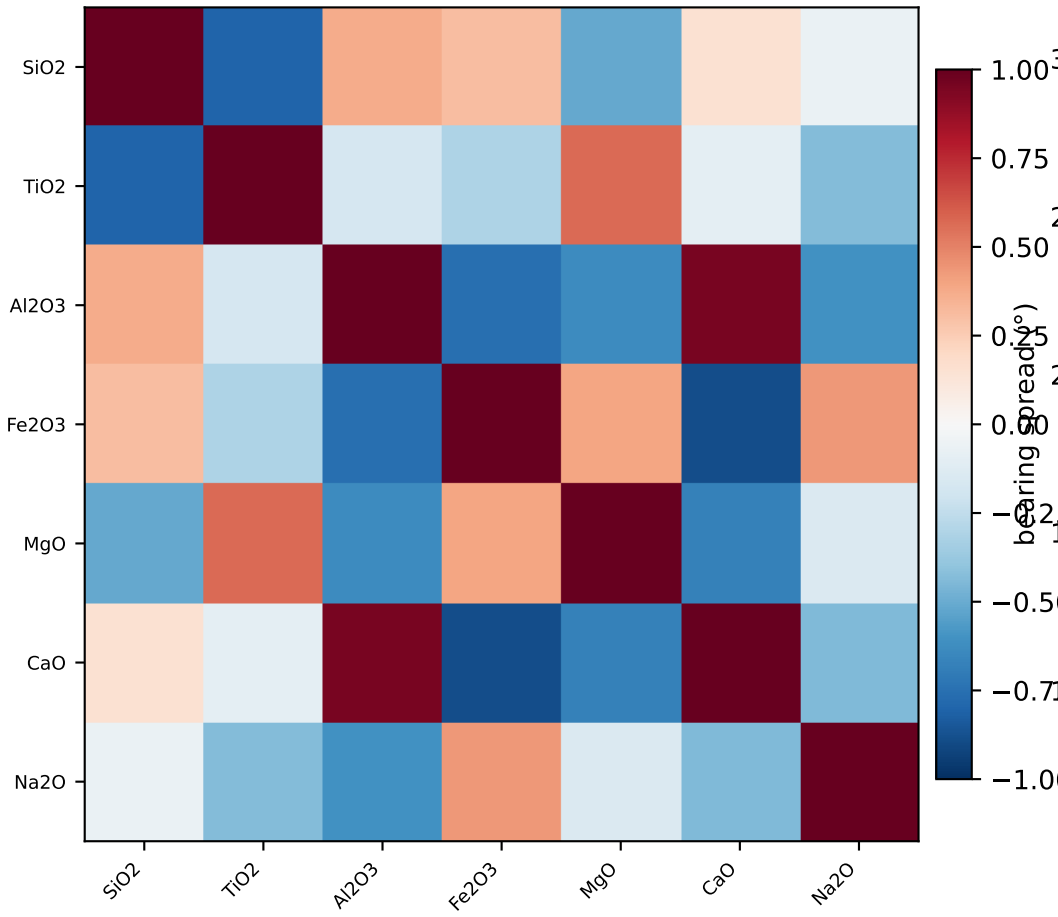
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

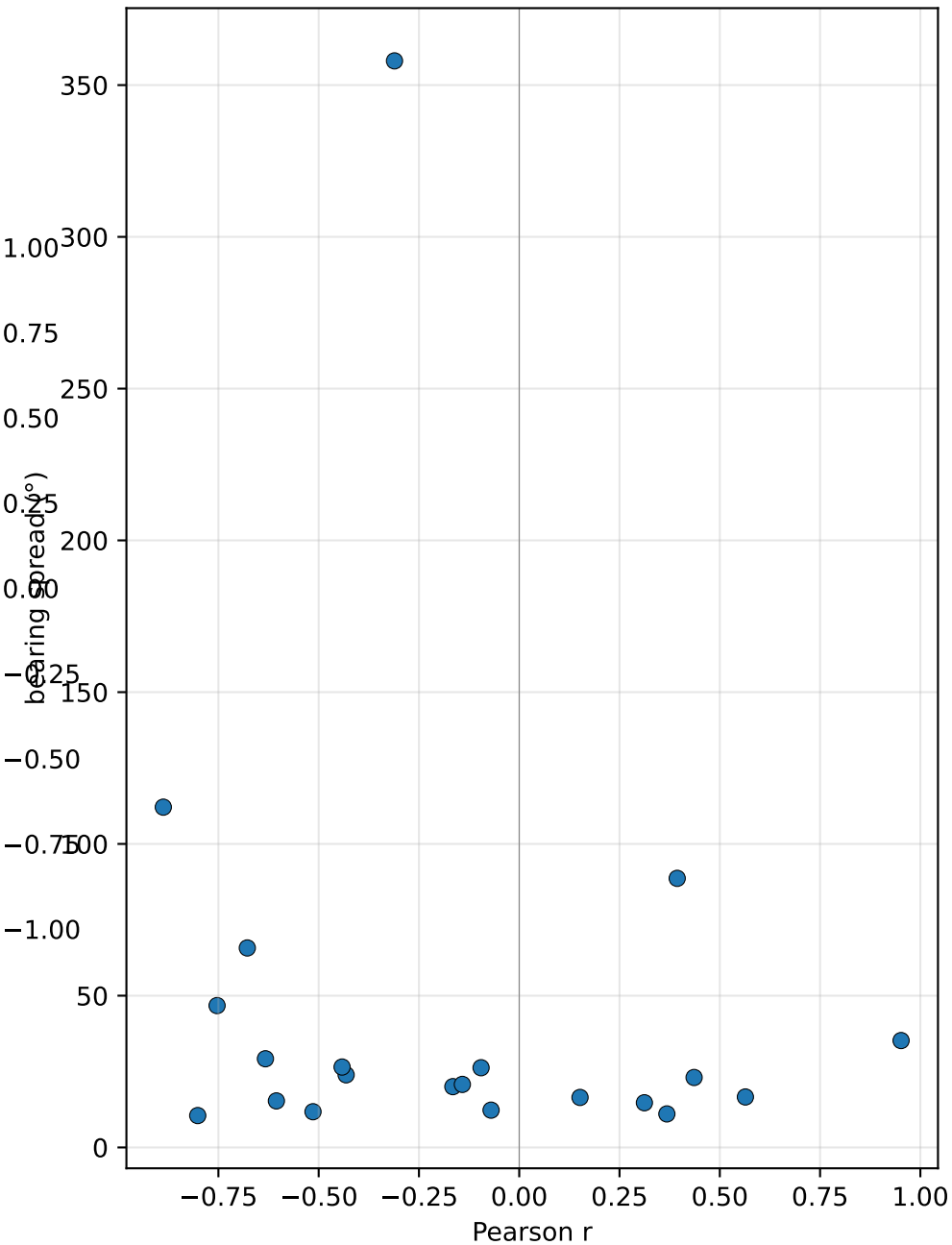


Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 7 | Pair Pearson r heatmap + co-movement scatter

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

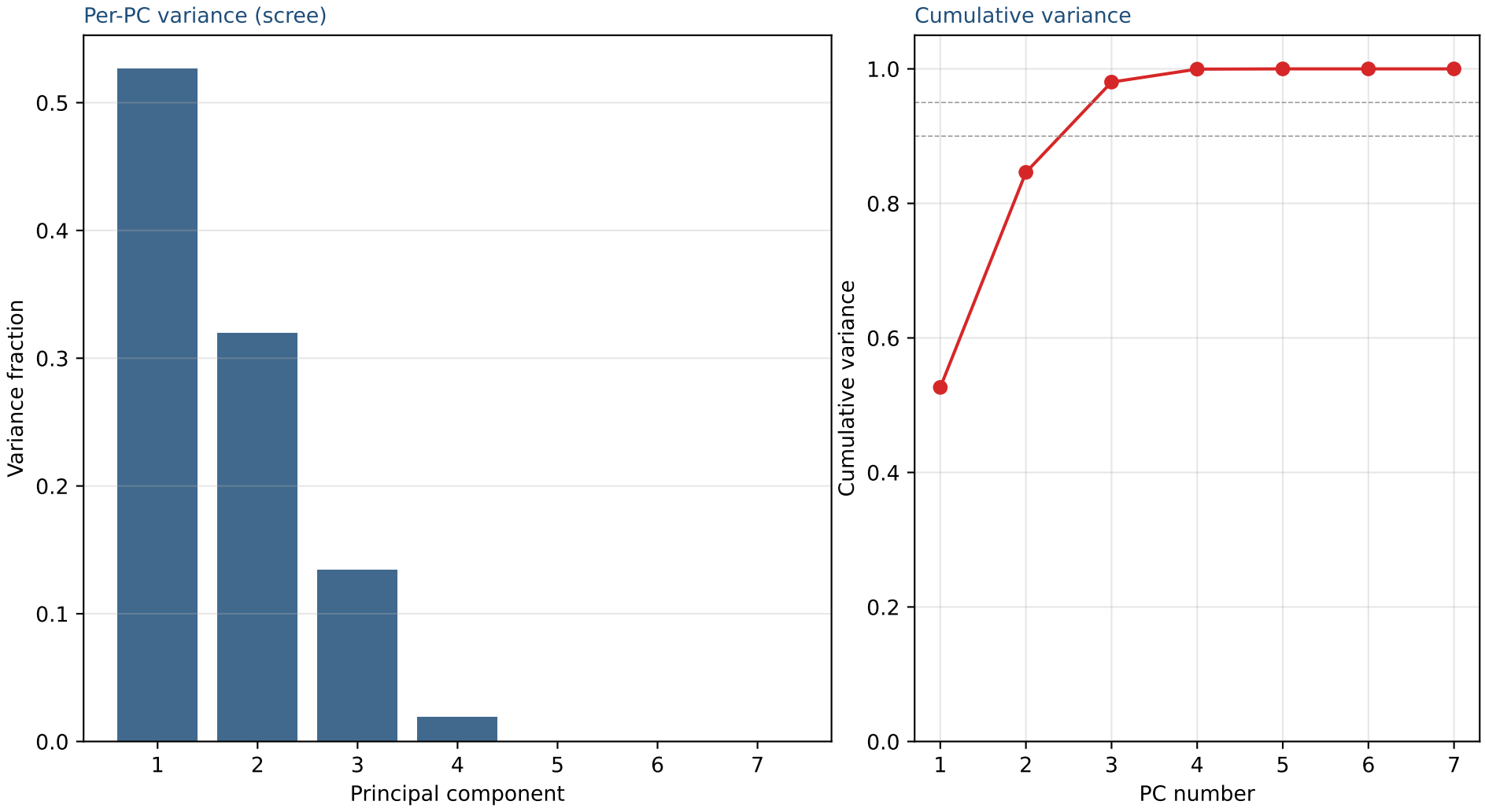




Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 8 | Scree Plot - CLR PCA

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2




```
Per-PC variance contribution:
  PC1: 52.65%   cum 52.65%
  PC2: 31.97%   cum 84.62%
  PC3: 13.42%   cum 98.03%
  PC4:  1.93%   cum 99.96%
  PC5:  0.04%   cum 100.00%
  PC6:  0.00%   cum 100.00%
  ... 1 more PCs
```

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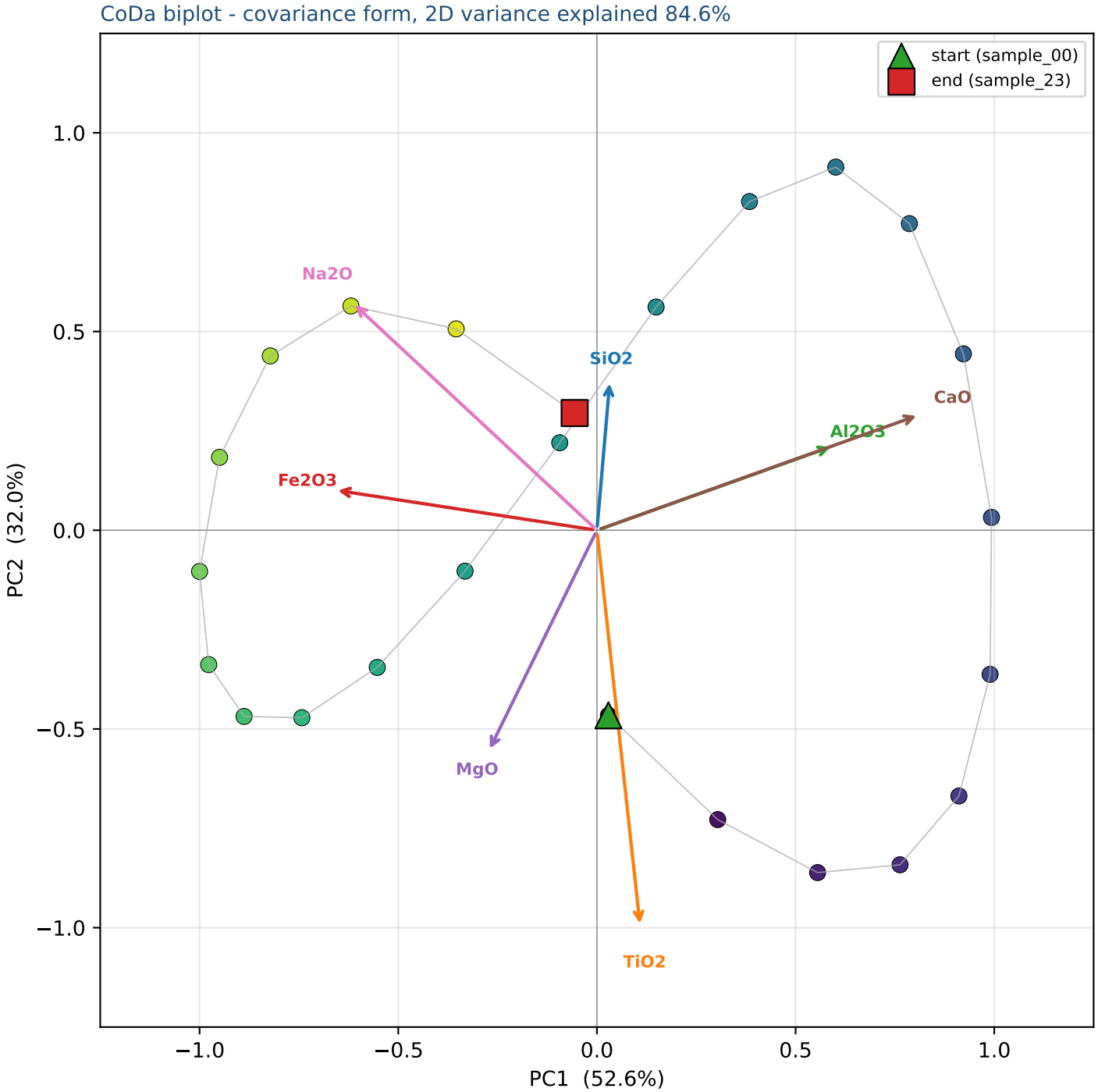
Plate 9 | CLR Biplot (CoDa biplot)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2




```
CoDa biplot (Aitchison-Greenacre):
  - Points: timestep scores in PC(1,2) of c
  - Rays:   carrier loadings (columns of V)
  - Ray length = carrier importance on the
  - Angle between rays ~ negative log-ratio
  - Distance between point projections = ap
    of the Aitchison distance restricted to

PC1 captures 52.6% of variance.
PC2 captures 32.0% of variance.
Total 2D plane:  84.6%
```

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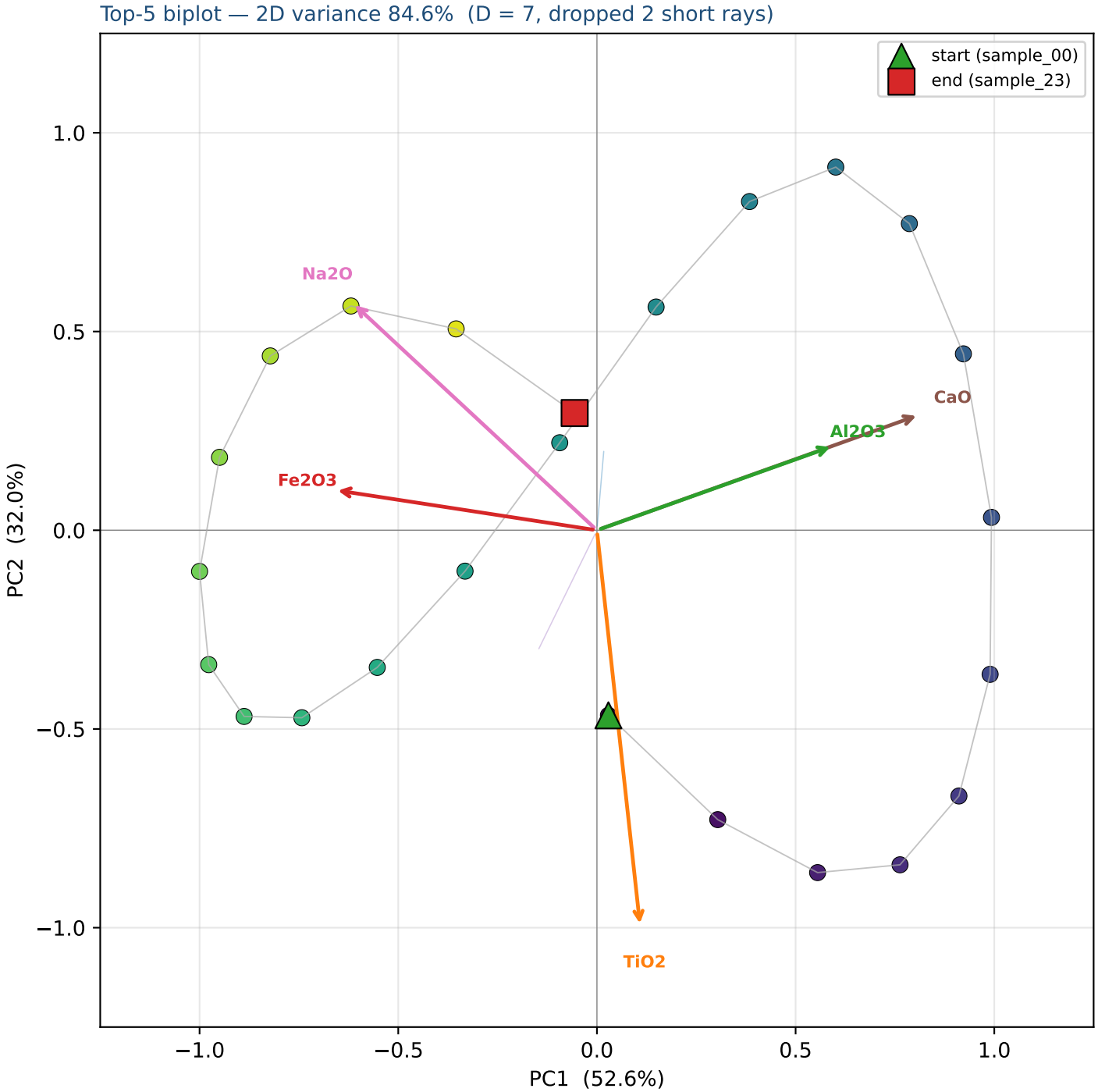
Plate 10 | CLR Biplot — top-5 carriers by loading magnitude

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Loading magnitudes $\sqrt{(G_1^2 + G_2^2)}$:

+ TiO2	0.7247
+ CaO	0.6228
+ Na2O	0.6068
+ Fe2O3	0.4828
+ Al2O3	0.4569

Dropped (faint stubs only):

MgO	0.4487
SiO2	0.2747

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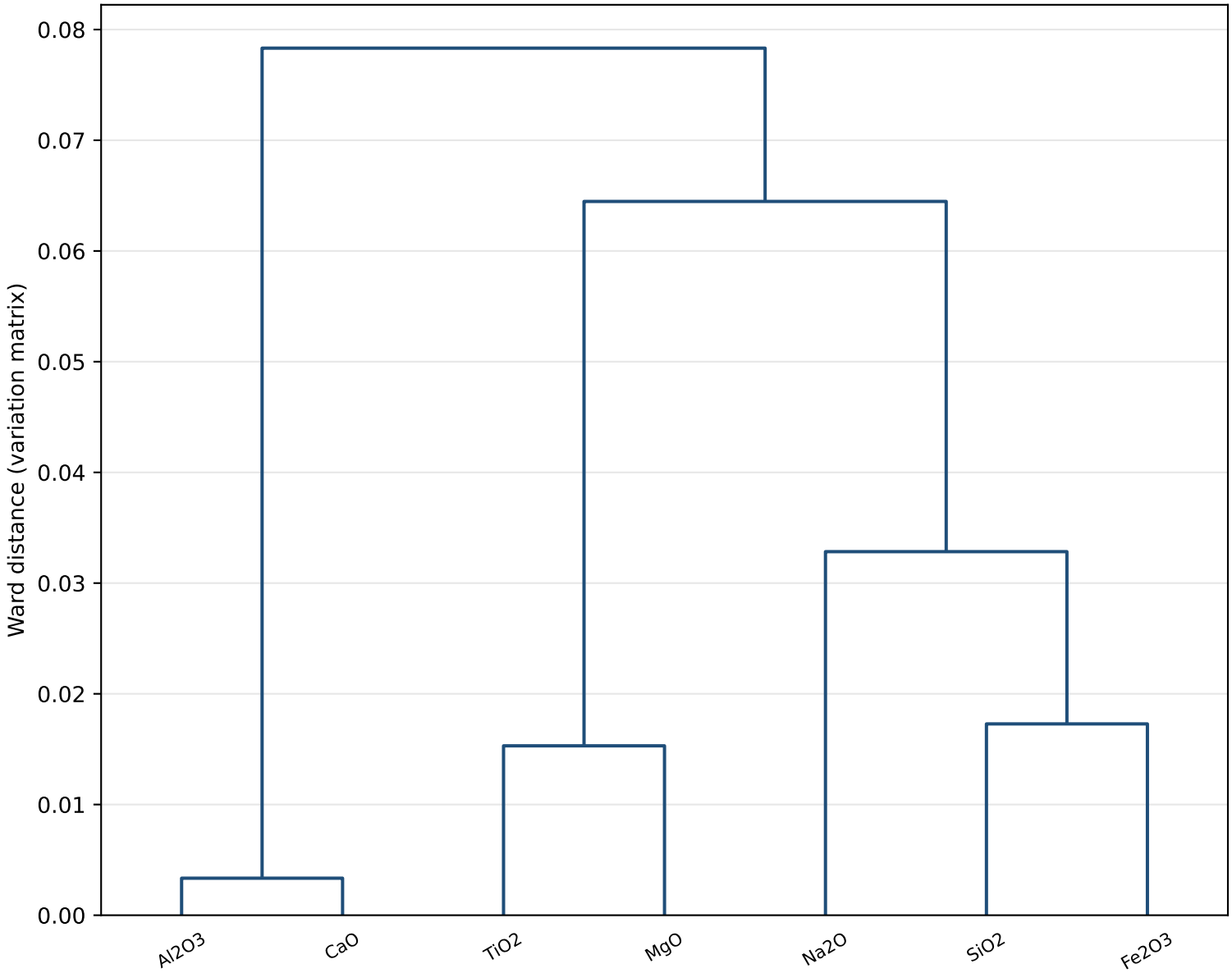
Plate 11 | Balance-Dendrogram (Ward on variation matrix)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Hierarchical balance tree — each merge defines one balance partition




```
Balance-dendrogram (Pawlowsky-Glahn-
  – Ward clustering on the (symmetri
  – Each internal node merges two ca
  – That merge defines a balance:
      b = sqrt(|A||B|/(|A|+|B|)) ·
    where g() is the geometric mean.
  – The balances form an orthonormal

D = 7 carriers → D-1 = 6 balances.
Tightly-clustered carriers (low vari
See companion plate for the full SBP
```

Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 12 | Sequential Binary Partition (SBP) Matrix

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

SBP cells: + = numerator, - = denominator, . = not in this balance

	b1	b2	b3	b4	b5	b6
SiO2	.	.	+	-	-	-
TiO2	.	+	.	.	+	-
Al2O3	+	.	.	.	.	+
Fe2O3	.	.	-	-	-	-
MgO	.	-	.	.	+	-
CaO	-	.	.	.	.	+
Na2O	.	.	.	+	-	-

Balance b_k = sqrt(|A_k||B_k|/(|A_k|+|B_k|)) * ln(g(A_k) / g(B_k)) | D = 7; D-1 = 6 balances.

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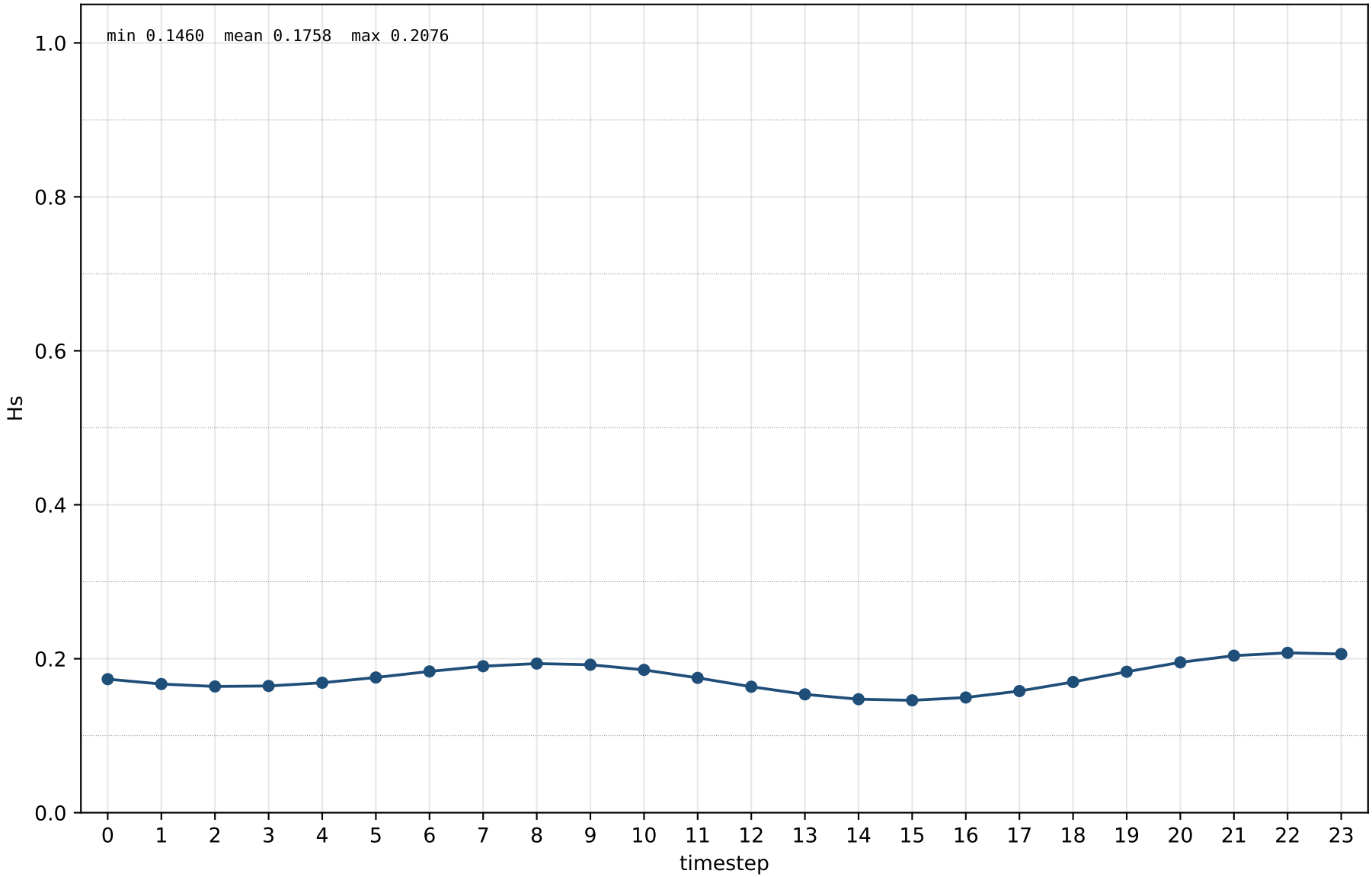
Plate 13 | Hs (Higgins scale) trajectory — Order 2 aggregate

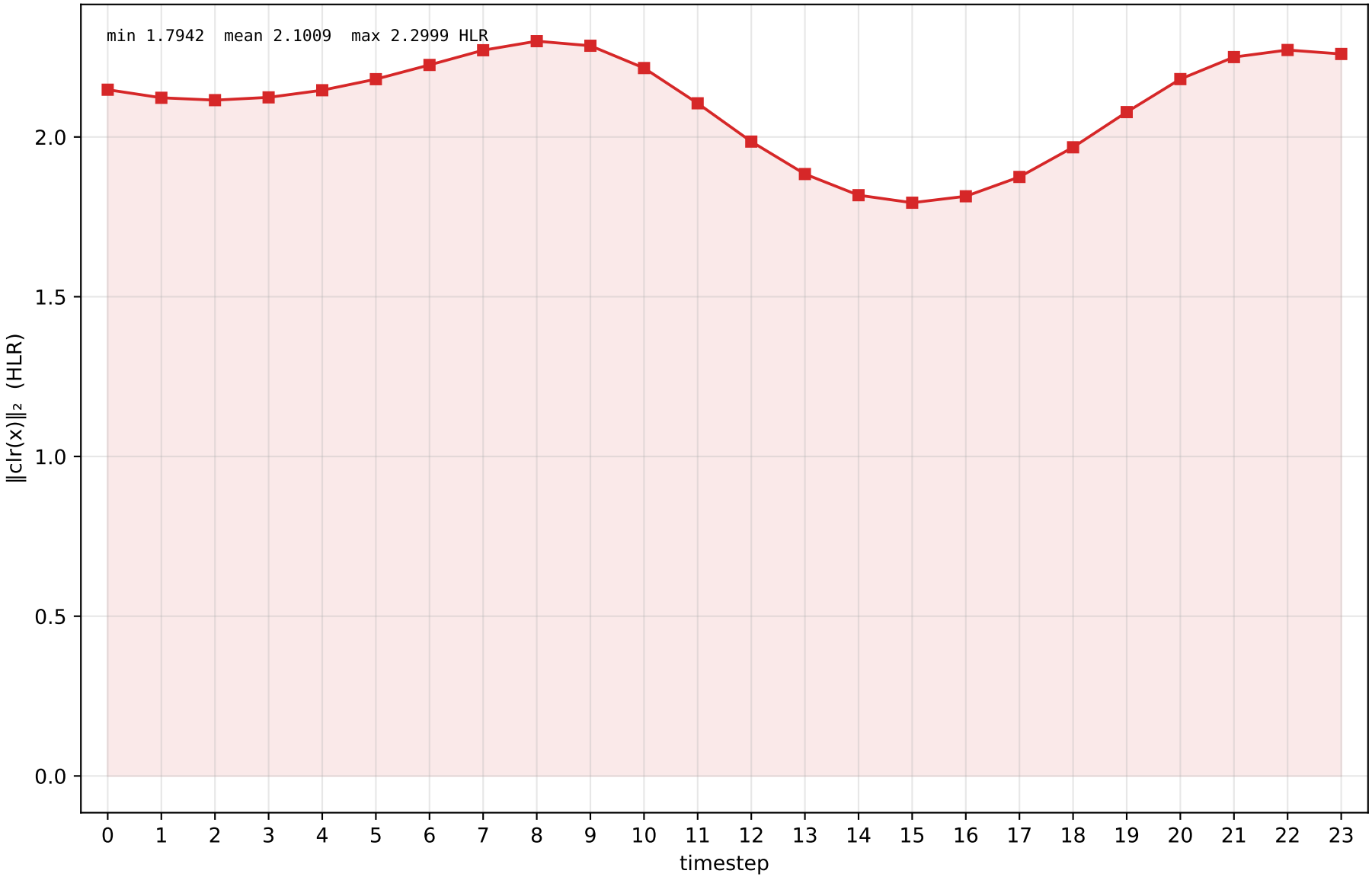
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 14 | Aitchison norm trajectory ‖clr(x)‖₂ — Order 2 aggregate

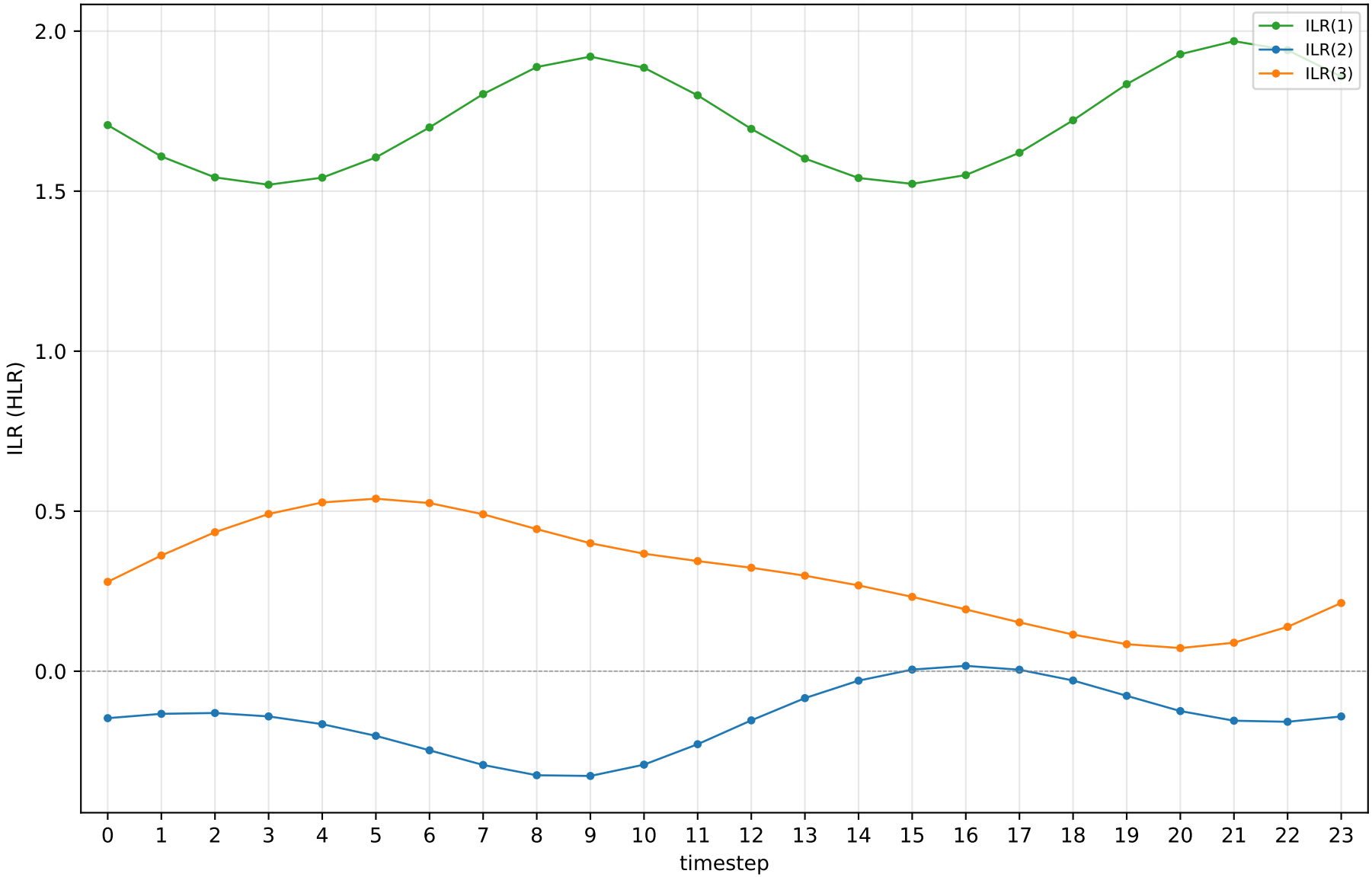
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 15 | ILR axes 1–3 trajectories (Helmert basis) — Order 2 aggregate

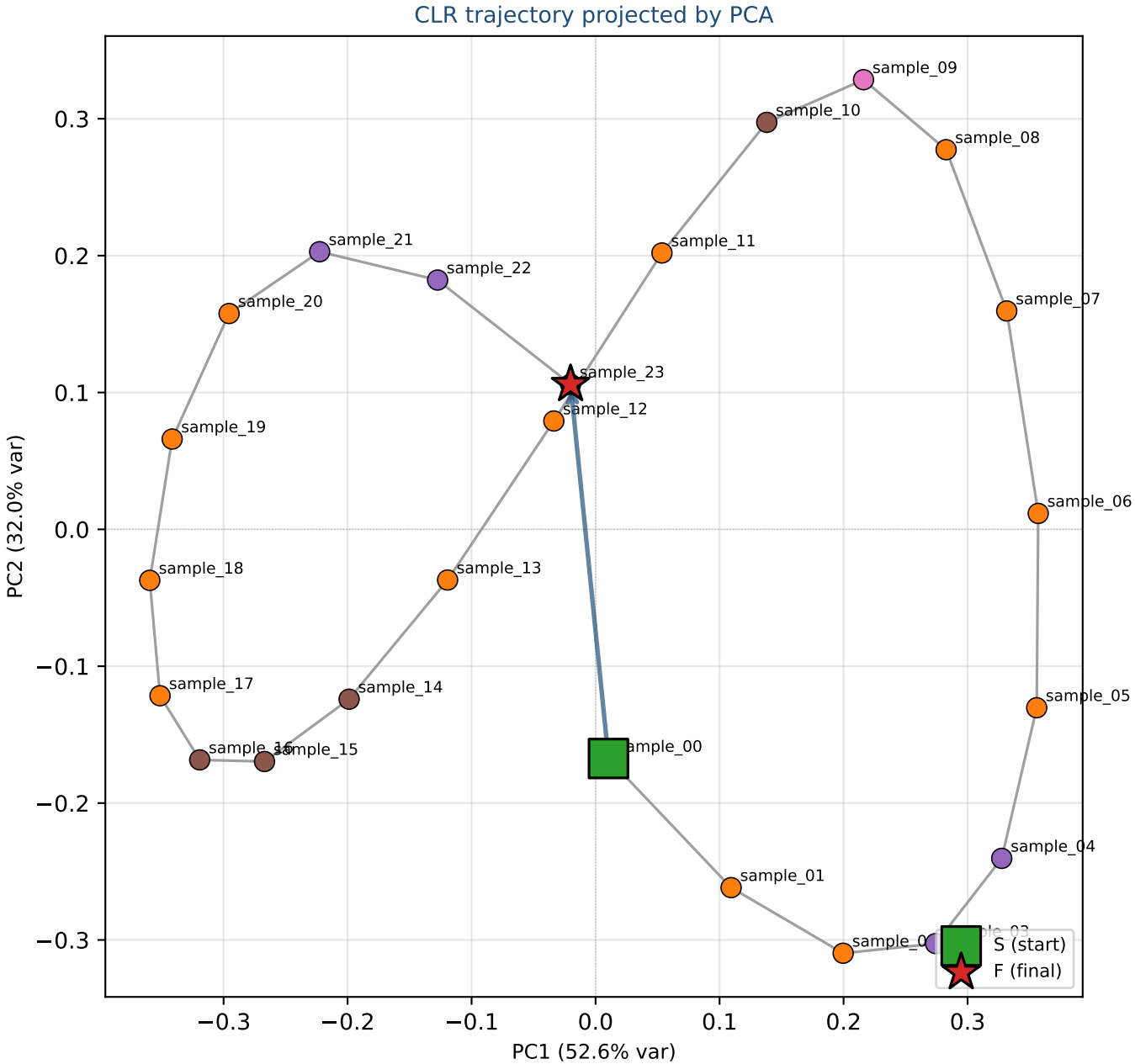
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 16 | System Course Plot — PCA 2D path with S→F net vector

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



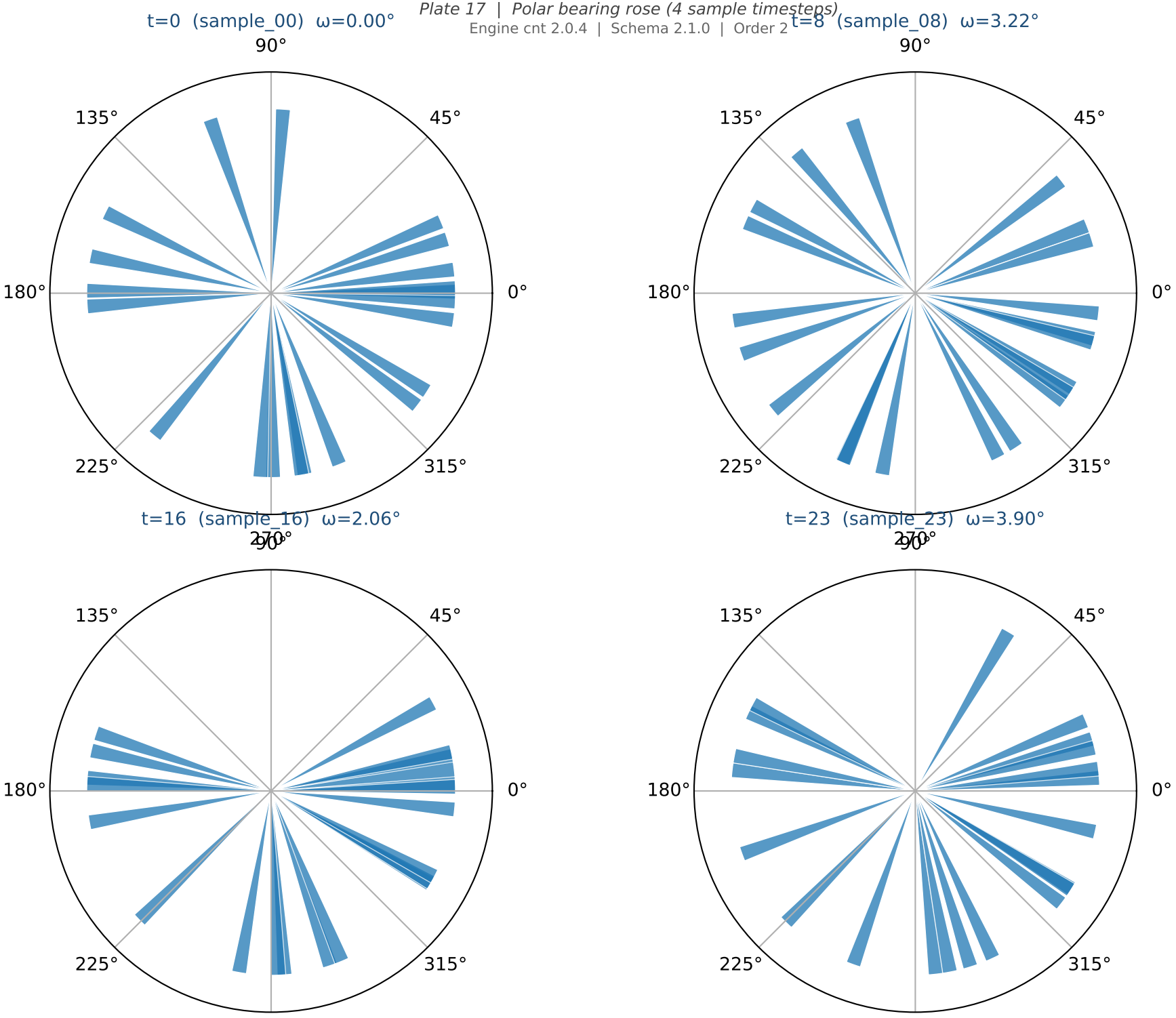
Navigation metrics (V_net = h_F − h_S)

```
h_start label    : sample_00
h_final label    : sample_23

net_distance     : 0.3722 HLR
path_length      : 2.7077 HLR
course_directness: 0.1375
  (1.0 = straight line, 0 = pure looping)

dynamic range S  : 2.6229 HLR
dynamic range F  : 2.6452 HLR

Reading guide:
  tight cluster  → stable composition
  long excursion → structural shift
  sharp turn     → regime transition
  loop           → cyclic behaviour
  spiral         → systematic drift
```

Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 17 | Polar bearing rose (4 sample timesteps)

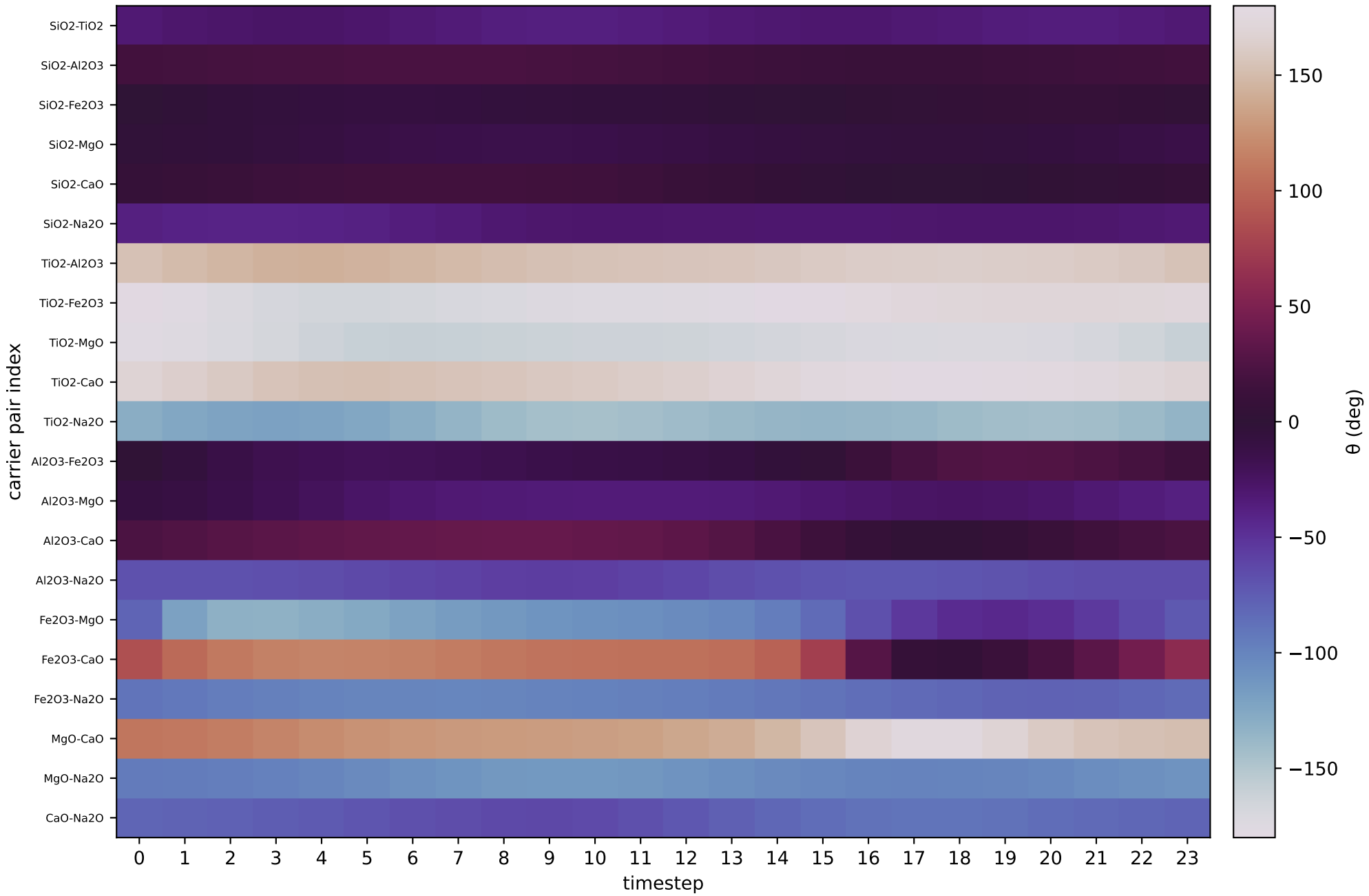
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 18 | Bearing heatmap T × C(D,2) pairs

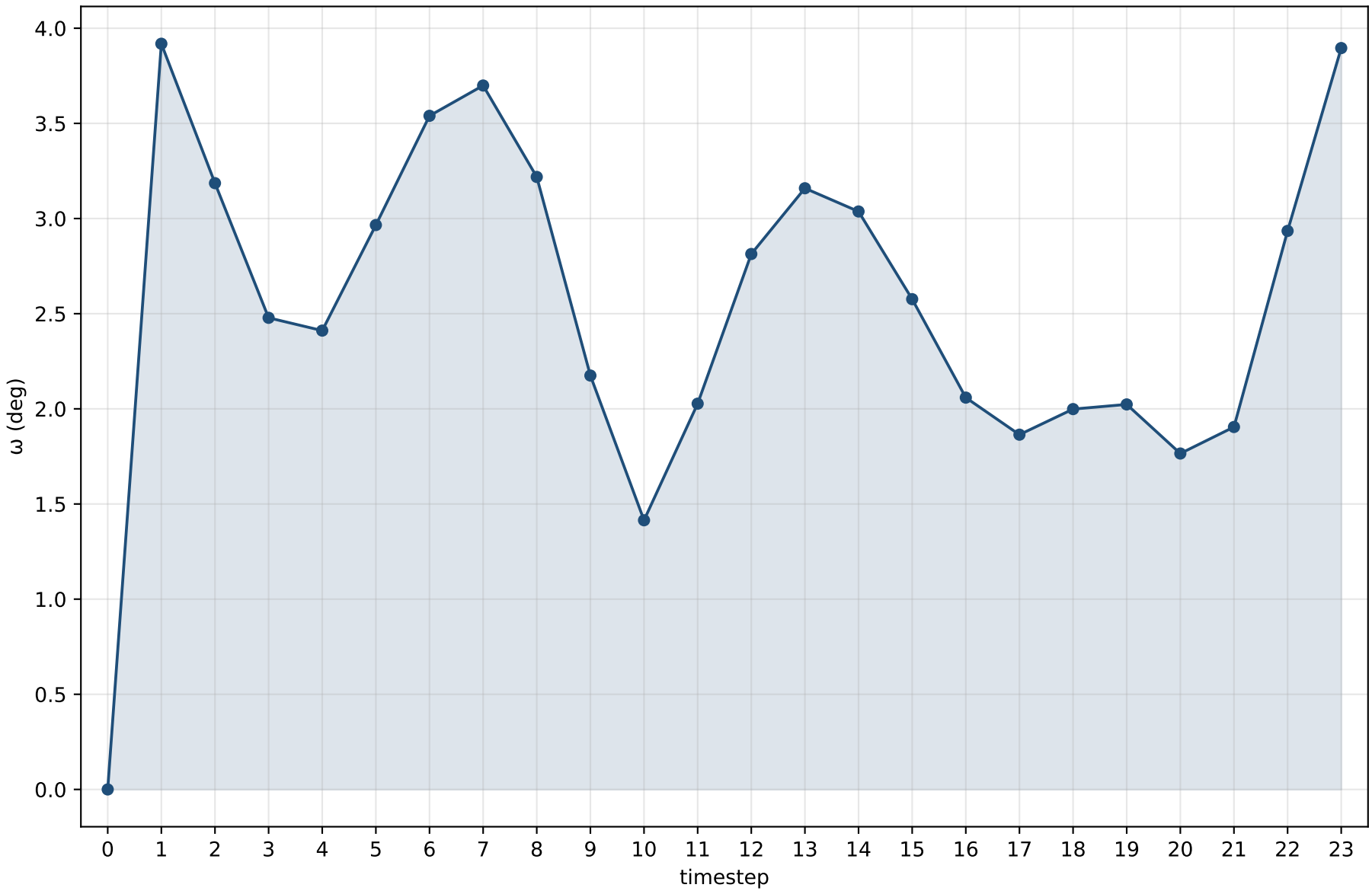
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 19 | ω angular velocity trajectory

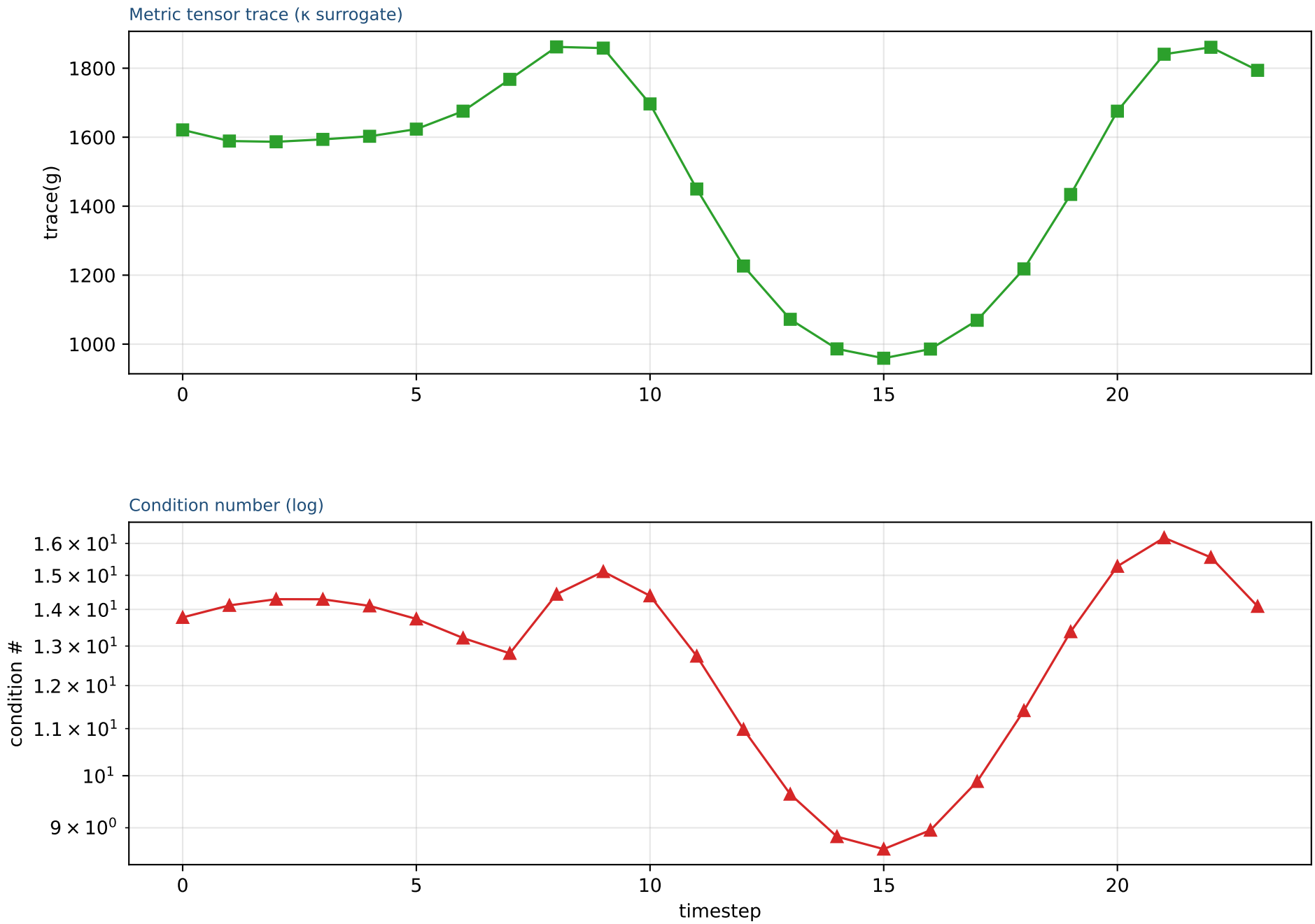
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 20 | Metric tensor trace + condition number

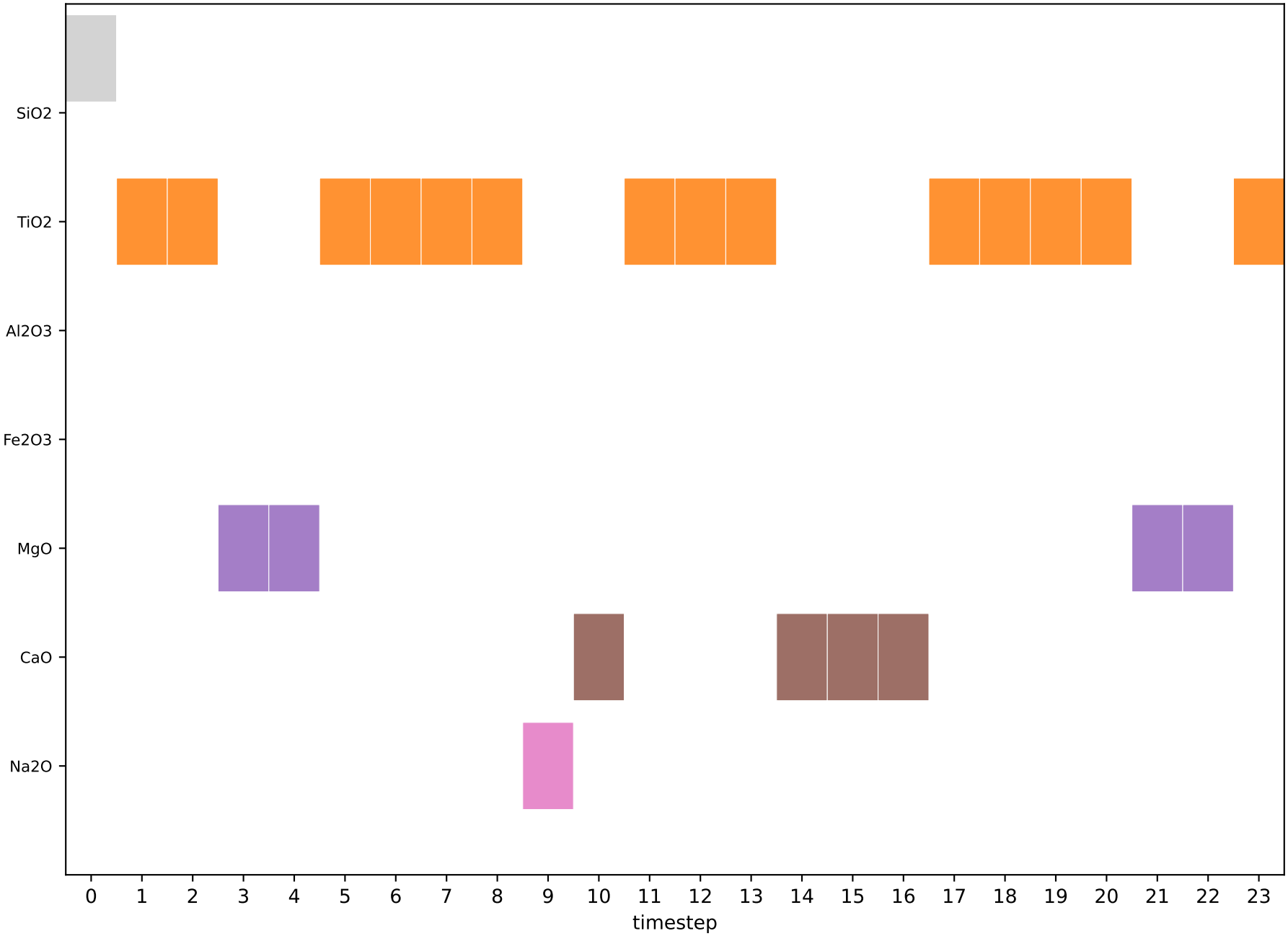
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 21 | Helmsman σ lineage timeline

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

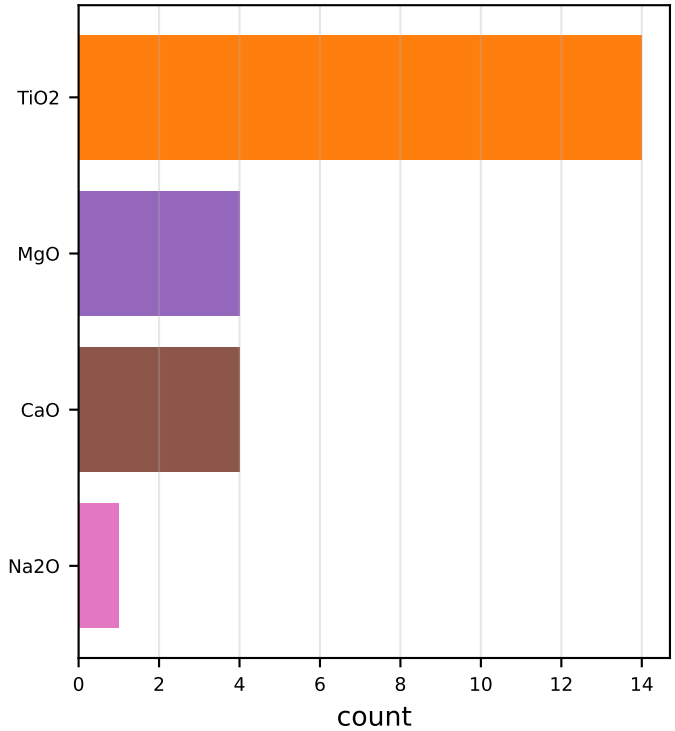


Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

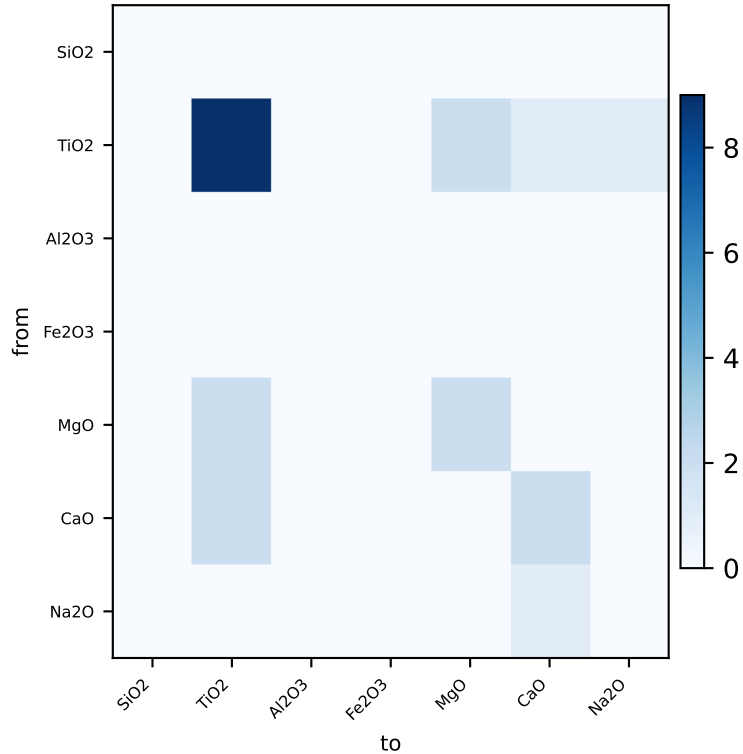
Plate 22 | Helmsman σ — frequency · transitions · per-year displacement

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

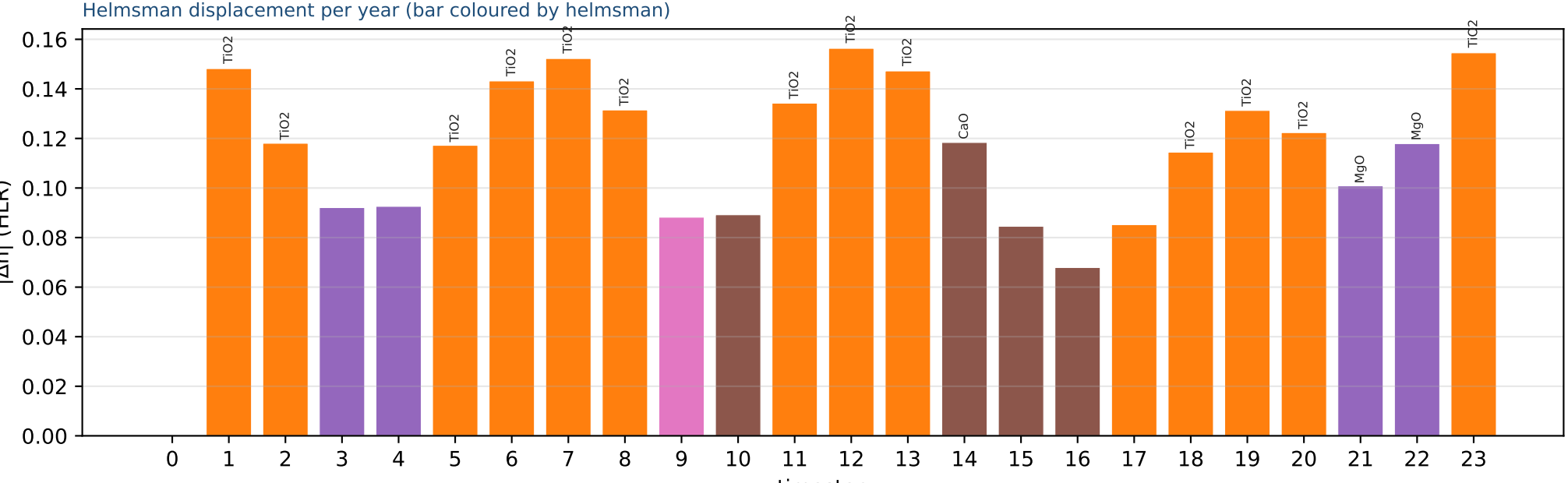
Helmsman frequency



Transition matrix



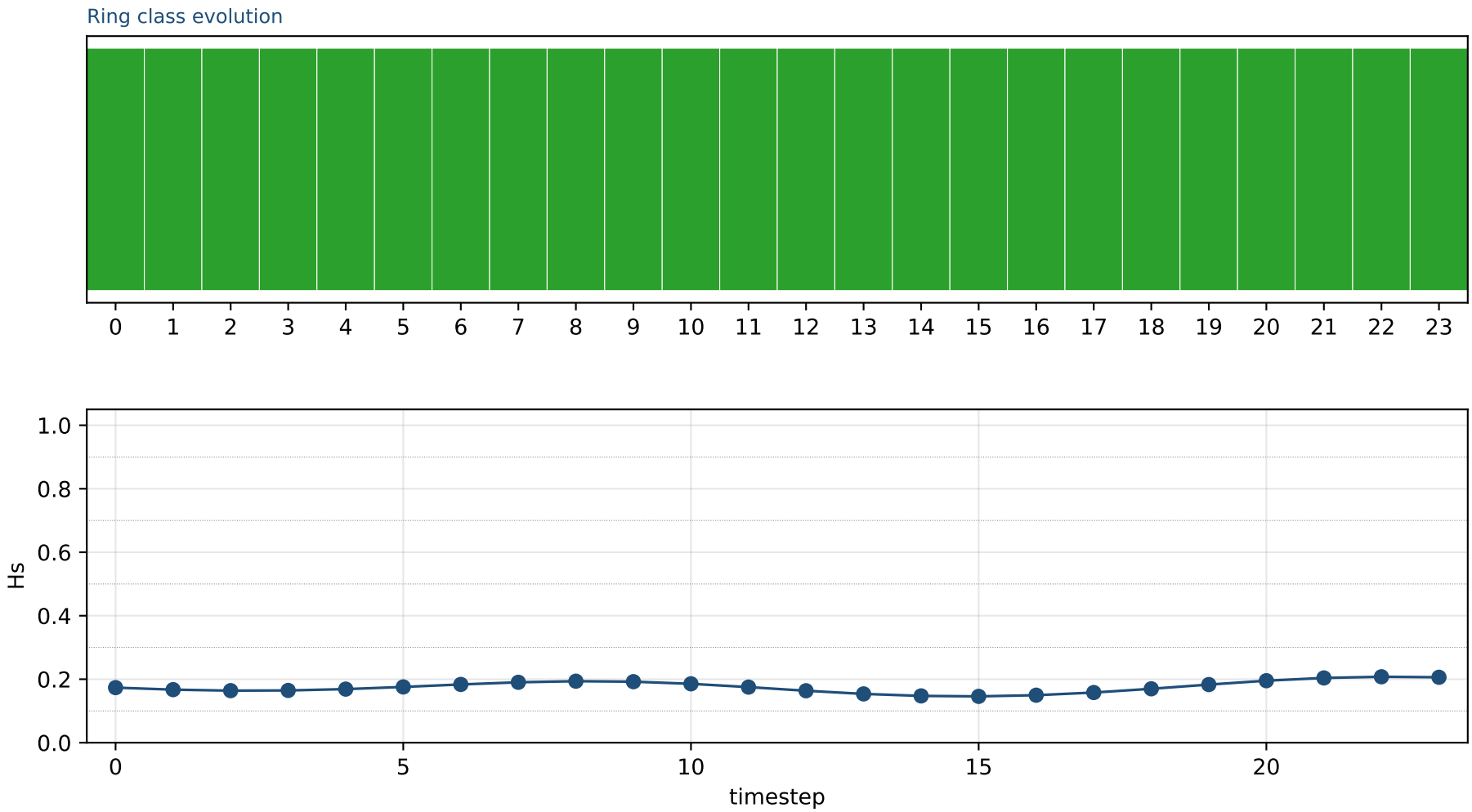
Helmsman displacement per year (bar coloured by helmsman)



Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 23 | Ring class evolution + Hs

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 24 | Lock event timeline (carriers × timesteps)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

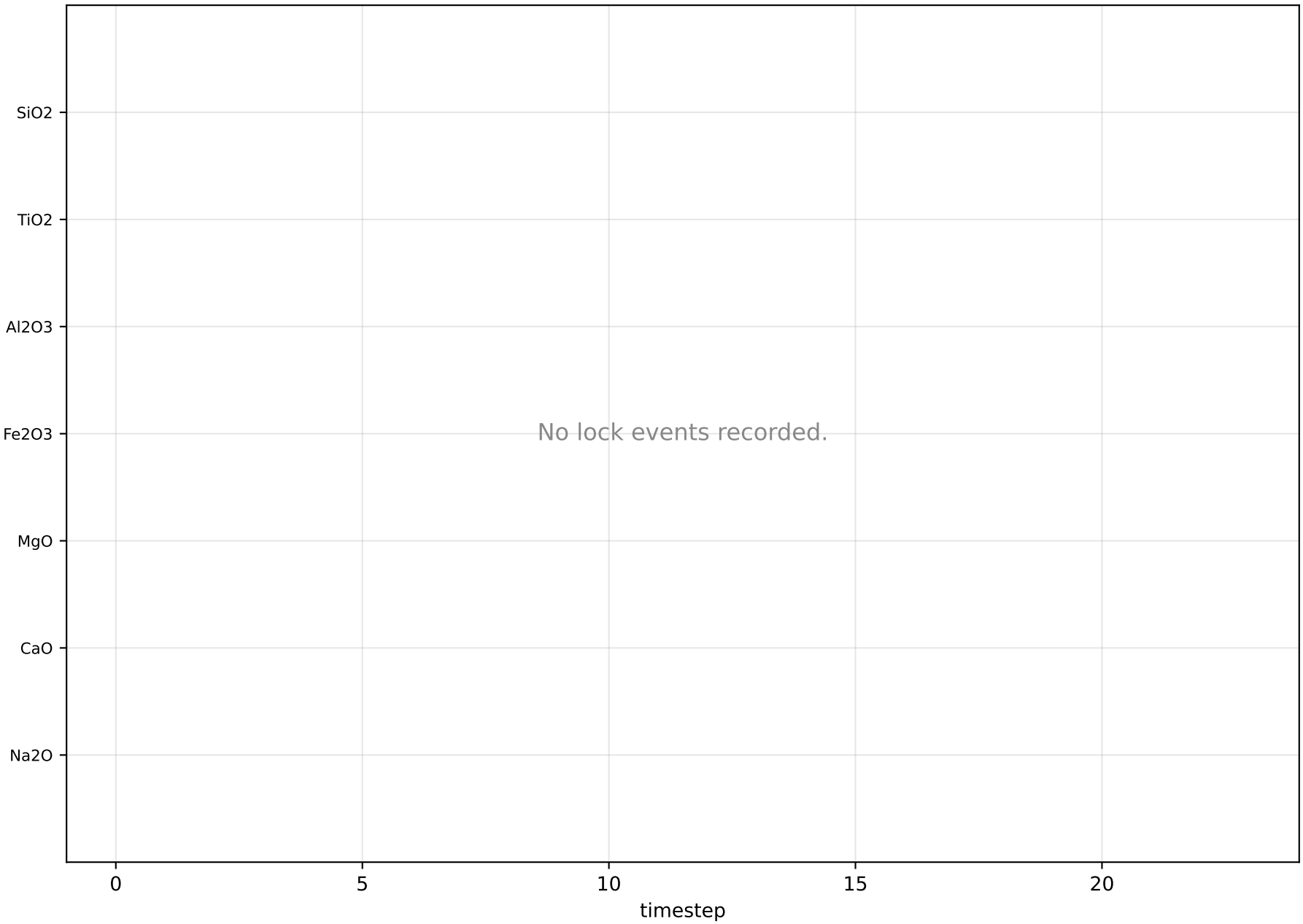


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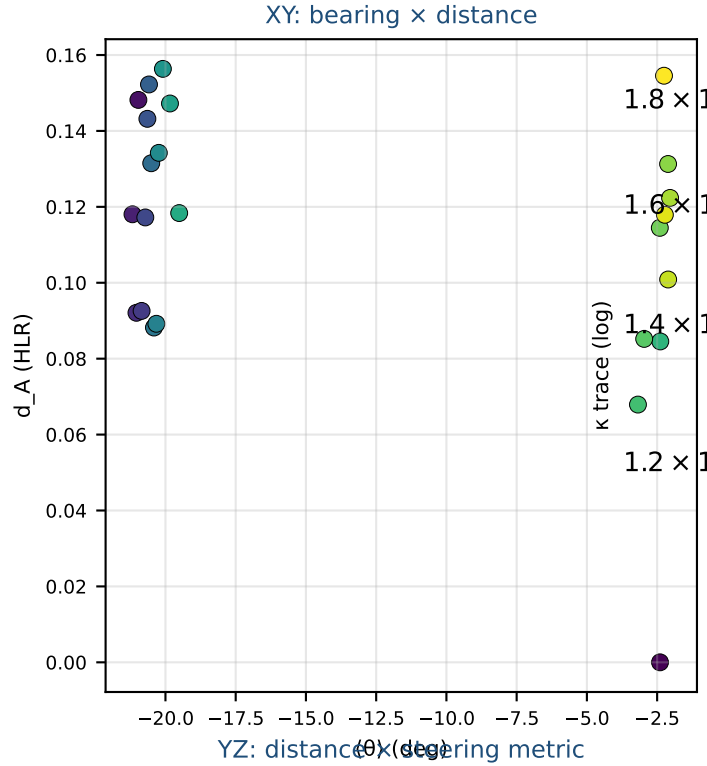
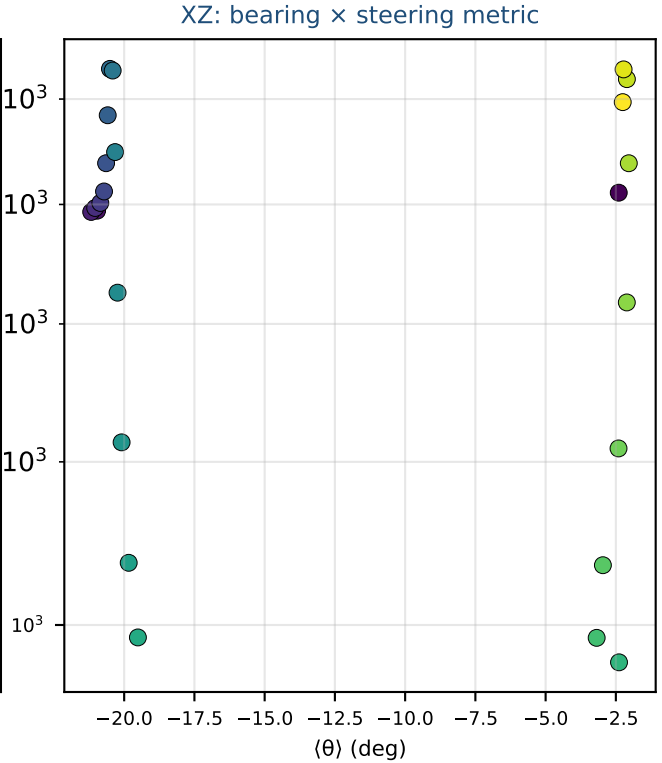
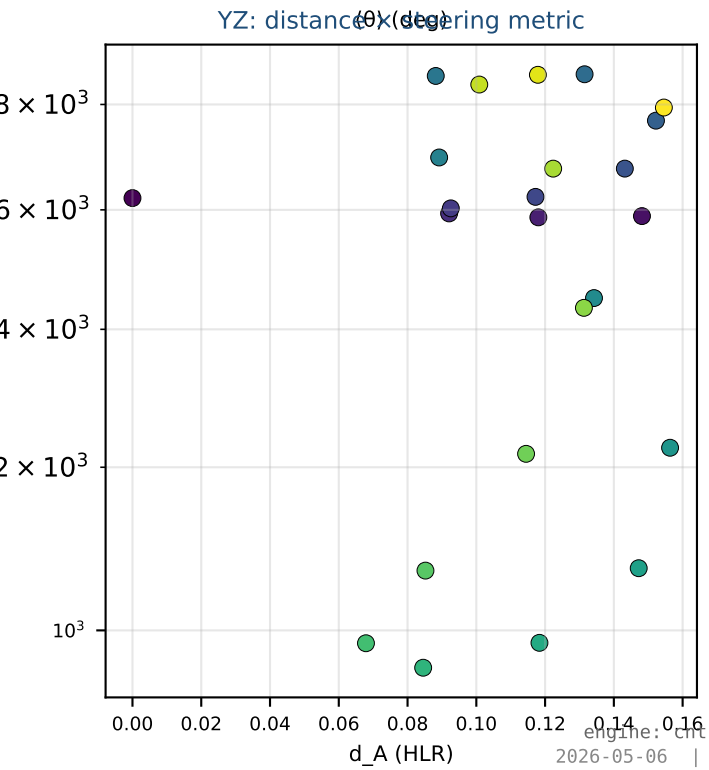
Plate 25 | Compositional Bearing Scope (CBS) — 3 orthogonal faces (Math Handbook Ch 16)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2







CBS info panel (Math Handbook Ch 16)

```
⟨θ⟩ range  : [-21.2, -2.0]°
d_A range  : [0.0000, 0.1564] HLR
κ range    : [9.59e+02, 1.86e+03]
Hs range   : [0.1460, 0.2076]
ω max      : 3.92°

Each face: one observation per timestep.
Colour = timestep (early=dark, late=bright).
Three faces locate the system in the
(bearing, distance, steering) measurement cube.
```

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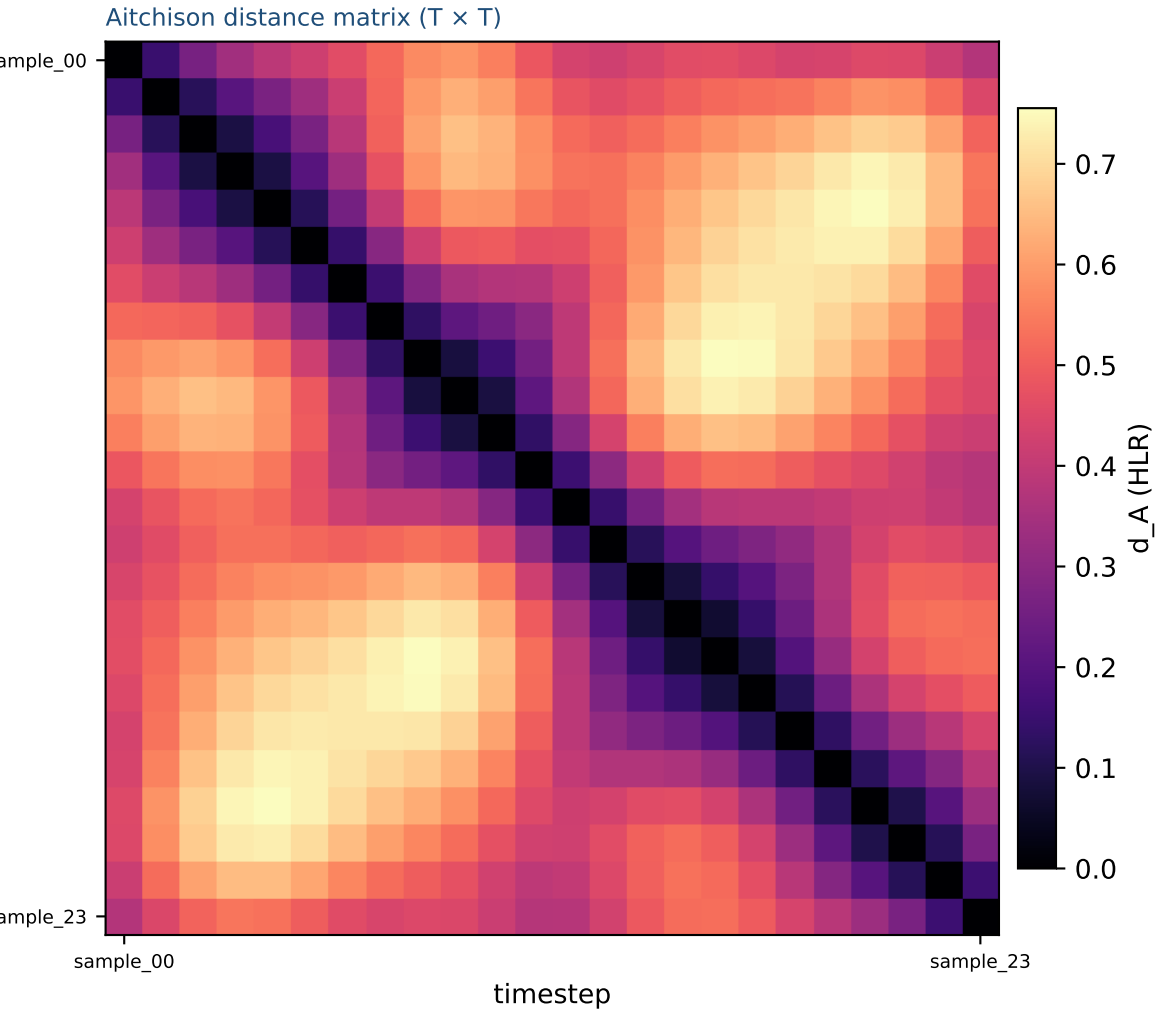
Plate 26 | Pairwise divergence — distance matrix + top-15 ranking

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

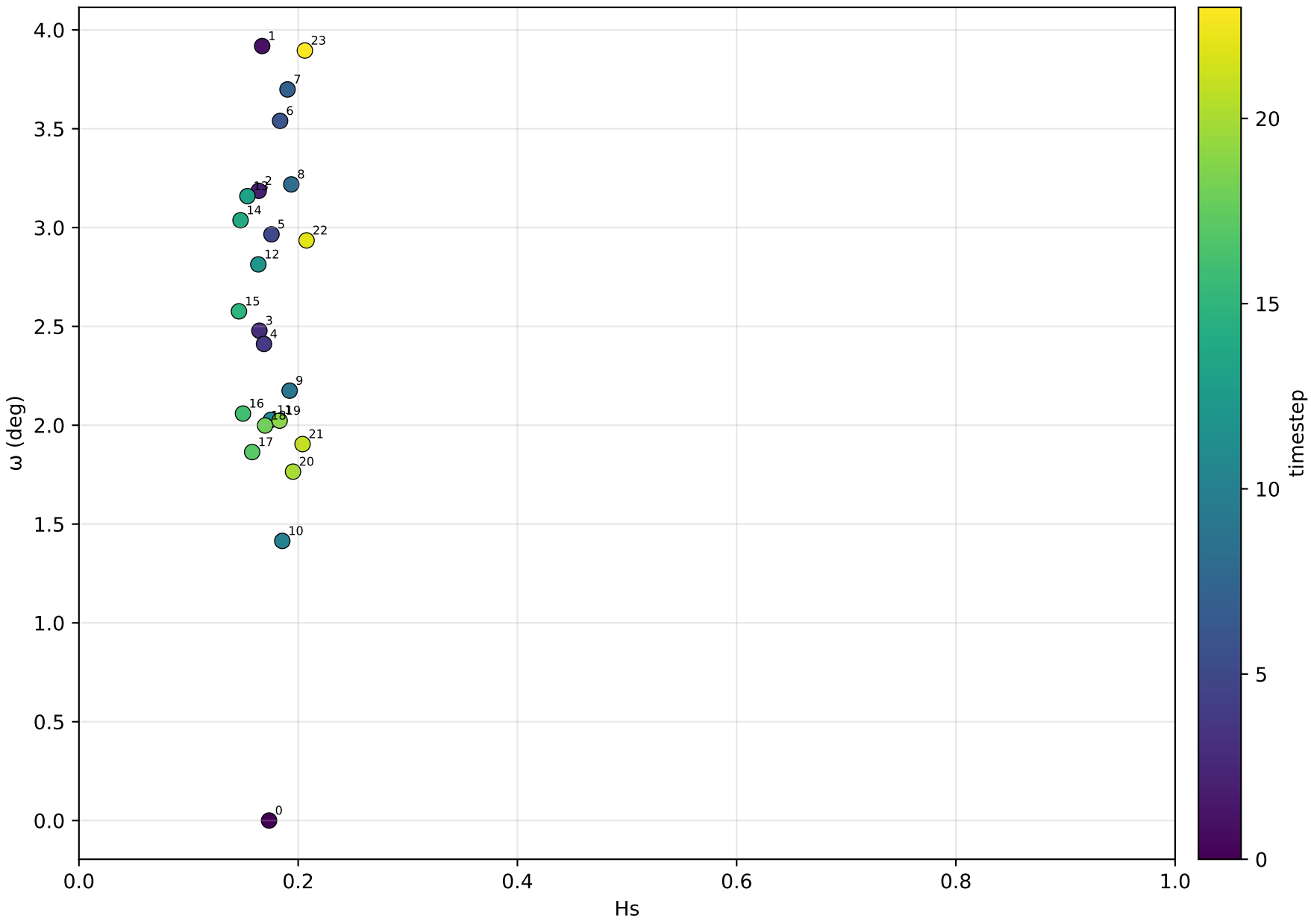


rank	A	B	d_A
1	sample_04	sample_20	0.7553
2	sample_08	sample_16	0.7545
3	sample_08	sample_17	0.7497
4	sample_04	sample_19	0.7424
5	sample_03	sample_20	0.7411
6	sample_07	sample_17	0.7390
7	sample_09	sample_16	0.7360
8	sample_05	sample_20	0.7352
9	sample_05	sample_19	0.7351
10	sample_07	sample_16	0.7342
11	sample_04	sample_21	0.7300
12	sample_09	sample_17	0.7258
13	sample_05	sample_18	0.7241
14	sample_03	sample_21	0.7236
15	sample_06	sample_17	0.7222

Compositional system geometry and dynamics for, ChemixHub HOIP-7 oxide samples sample_00 → sample_23

Plate 27 | ω vs Hs scatter (XY test)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



```
T = 24,  D = 7
carriers: SiO2, TiO2, Al2O3, Fe2O3, MgO, CaO, Na2O

Navigation (System Course Plot):
  net_distance     = 0.3722 HLR
  path_length      = 2.7077 HLR
  course_directness= 0.1375

ω (deg)        : min 0.00  mean 2.54  max 3.92
Hs             : min 0.1460  mean 0.1758  max 0.2076
Aitchison norm : min 1.7942  mean 2.1009  max 2.2999
κ trace        : min 959.13  max 1861.77
condition #    : min 8.62e+00  max 1.62e+01

Bearings: range [-178.74, 179.27]°,  std 96.46°
Lock events    : 0
Pairs          : 21
|r| > 0.7 pairs: 4

Helmsman frequency (top 5):
  TiO2            14  (58.3%)
  MgO              4  (16.7%)
  CaO              4  (16.7%)
  (none)           1  (4.2%)
  Na2O             1  (4.2%)

Variation matrix trace : 0.0000
```